

A Confidence-Gated Agent for Query-Organisation Relevance: Routing Hard Cases from Fine-Tuned RuModernBERT to a Search-Augmented LLM

Anastasia Chernaya

June 2026

Contents

1	Introduction	2
1.1	Contributions	2
2	Dataset	3
2.1	Source and Provenance	3
2.2	Binarisation: Why Class 0.1 Is Excluded	3
2.3	Splits	3
2.4	Missing Values	3
2.5	Text Construction at Runtime	4
2.6	Dataset Statistics	4
3	System Architecture	4
3.1	Routing Logic	4
3.2	Training Configuration	5
3.3	Temperature Scaling	6
4	Pipeline Stages	6
4.1	Stage 0 — EDA	6
4.2	Stage 1 — TF-IDF Baseline	6
4.3	Stage 2 — Fine-tune RuModernBERT + Temperature Scaling	6
4.4	Stage 3 — Error Analysis and Routing Threshold	6
4.5	Stage 4 — Hybrid System on Val	7
4.6	Stage 5 — Final Evaluation on Eval (Locked)	7
5	Metrics	7
5.1	Accuracy	7
5.2	Macro-F1	7
6	Results	8
6.1	Overall Performance	8
6.2	Routing and LLM Agent on Eval	8
6.3	Error Matrix: Hybrid vs. BERT-only	8
7	Known Limitations	9

Abstract

We address binary relevance classification of organisations to broad user queries on Yandex Maps. A fine-tuned RuModernBERT-base cross-encoder serves as the main classifier; a confidence routing threshold directs low-certainty examples to an LLM agent (LangGraph + VseGPT) that may additionally call a web-search tool. The routing threshold and the decision whether to enable web search are both derived from a quantitative error taxonomy of classifier mistakes, not assumed a priori. On the held-out evaluation set the hybrid system achieves **78.2%** accuracy and **0.781** macro-F1, an improvement of 1.1 pp in accuracy over the BERT-only baseline and 17 pp over TF-IDF + Logistic Regression. The project code is available at github.com/ChernayaAnastasia/RuModernBERT-llm-router.

1 Introduction

Modern map and local-search services answer hundreds of millions of queries such as “*шиномонтаж 24 часа*” or “*кафе с животными в москве*”. For every query the system must decide which of the surrounding organisations are relevant and should be shown to the user. The quality of this decision directly shapes the product: irrelevant results erode user trust and increase abandonment, while relevant organisations that are wrongly filtered out lose visibility and customers. Because the task operates at the scale of an entire map service, even small improvements in relevance translate into a large absolute number of better answers, making the problem both practically important and economically significant.

Query-organisation relevance is a short-text, asymmetric matching problem. On one side there is a very short, often underspecified user query; on the other, a structured organisation listing containing a name, an address, a rubric (category), user reviews, and a price list. The difficulty is that relevance frequently depends on attributes not literally present in the listing text: whether the place is open around the clock, whether it carries a specific product, or whether it is suitable for children. Purely lexical methods cannot bridge this gap, and even a text-only neural classifier is limited to the information contained in the listing itself.

In this work we cast the task as binary classification (relevant vs. irrelevant) and propose a **confidence-gated hybrid cascade** instead of a single monolithic model. A compact fine-tuned transformer handles the bulk of traffic at very low cost; a confidence threshold routes only uncertain cases to a more capable but more expensive LLM agent that can call a web-search tool to retrieve external, real-world knowledge. Crucially, the decision of whether to add web search is not assumed in advance: we first build a taxonomy of residual errors and enable search only because a measurable share of those errors provably require external facts.

1.1 Contributions

- We prepare and characterise a binary query-organisation relevance dataset based on Yandex Maps data, including a principled binarisation of ambiguous (borderline) labels and a held-out out-of-distribution subset used as a sanity check.

- We establish two single-model baselines: TF-IDF + Logistic Regression and a fine-tuned RuModernBERT-base transformer.
- We propose a confidence-gated hybrid agent (BERT $\rightarrow$ LLM $\rightarrow$ optional web search) whose routing threshold and search-vs.-no-search decision are both grounded in a quantitative error analysis.
- We evaluate the system in a cost-aware setting, reporting accuracy, macro-F1, monetary cost of inference, and the measured contribution of web search to final quality.

2 Dataset

2.1 Source and Provenance

The [data](#) are provided by Yandex and represent the results of relevance annotation performed by human assessors. They are available for educational and research purposes only. The raw data are stored in JSONL format in `data/raw/`. After removing five noisy duplicates identified by the query + permalink key, the dataset contains **35 089** rows with original relevance labels $\{0, 0.1, 1\}$: class 0 accounts for 41%, class 0.1 for 13%, and class 1 for 46%.

2.2 Binarisation: Why Class 0.1 Is Excluded

A/B experiments on TF-IDF (`notebooks/eda.ipynb`) showed that class 0.1 is not separable by a linear model: $F1 \approx \text{random}$, uniform confusion with both neighbouring classes. Experiment C confirmed that fully excluding 0.1 yields the best binary validation accuracy (≈ 0.639 vs. ≈ 0.633 for strategies that remap $0.1 \rightarrow 0$ or $0.1 \rightarrow 1$). All subsequent stages use labels $\{0, 1\}$.

2.3 Splits

Splits are 70/15/15%, stratified by binary label, row-level.

Table 1: Dataset splits after binarisation.

Split	Rows	Irrelevant (0)	Relevant (1)	Note
train	21 270	10 152	11 118	
val	4 558	2 176	2 382	calibration & threshold tuning locked until Stage 5
eval	4 558	2 176	2 382	
OOD (rel_minus)	4 703	—	—	sanity check

2.4 Missing Values

Missing values in `COL_REVIEWS` ($\approx 4\%$) and `COL_PRICELIST` ($\approx 41\%$) are not dropped: the association between missingness and class label is statistically weak (Cramér’s $V < 0.1$). LLM-agent routing is also not based on field presence—accuracy when both review and price-list fields are empty is no worse than with full context ($\Delta \approx -0.02$ per EDA).

2.5 Text Construction at Runtime

COL_COMBINED_TEXT and COL_ORG_TEXT are *not* stored in parquet—they are built on load from `utils/data_loader.py`. The price list is included in `org_text` and in the LLM context—it matters for queries that mention a specific service or product. Cross-encoder tokenisation uses the scheme [CLS] query [SEP] `org_text` [SEP] with `truncation='only_second'` so the query is never truncated.

Data example.

Query: ремонт электроинструмента воронеж

Org text: Эвис-Моторс центр; Evis-Motors Central; Evis-Motors-Central; Эвис Моторс; Эвис-Моторс; Evis-Motors | Москва, Партийный переулок, 1к58с1 | Автосервис, автотехцентр | Организация занимается автообслуживанием, включая замену масла, диагностику, кузовной ремонт и хранение шин. Отзывы в основном положительные, но есть и критика. Хвалят: профессионализм мастеров, качество обслуживания, вежливость персонала. Критикуют: длительное ожидание, завышенные цены, невнимательность к клиентам.

Label: 0 (not relevant)

2.6 Dataset Statistics

Table 2: Token and document statistics per split.

	Train	Val	Eval
Documents	21 270	4 558	4 558
Irrelevant (0)	10 152	2 176	2 176
Relevant (1)	11 118	2 382	2 382
Vocabulary size	192 706	80 829	80 718
Total tokens	6 003 232	1 282 737	1 281 283

3 System Architecture

The hybrid system has two components.

BERT cross-encoder. `deepvk/RuModernBERT-base` [1] is fine-tuned as a binary classifier on pairs [query, `org_text`] [2]. It processes all examples and outputs calibrated probabilities via temperature scaling [3].

LLM agent. The agent/ module (LangGraph + VseGPT) is invoked only for low-confidence examples, optionally using Tavily web search [4]. The decision of whether to call Tavily is derived from a manual error taxonomy (Section 4.4).

3.1 Routing Logic

Let p_1 denote the calibrated probability that an example belongs to class 1. Model confidence is defined as:

$$\text{bert_max_proba} = \max(p_1, 1 - p_1).$$

If $\text{bert_max_proba} \geq \tau$ (confidence threshold τ), the example is classified by BERT directly (routed_to = bert). Otherwise, it is forwarded to the LLM agent, which may invoke a web-search node before producing a final prediction (routed_to = llm).

The full routing pipeline is shown in Figure 1.

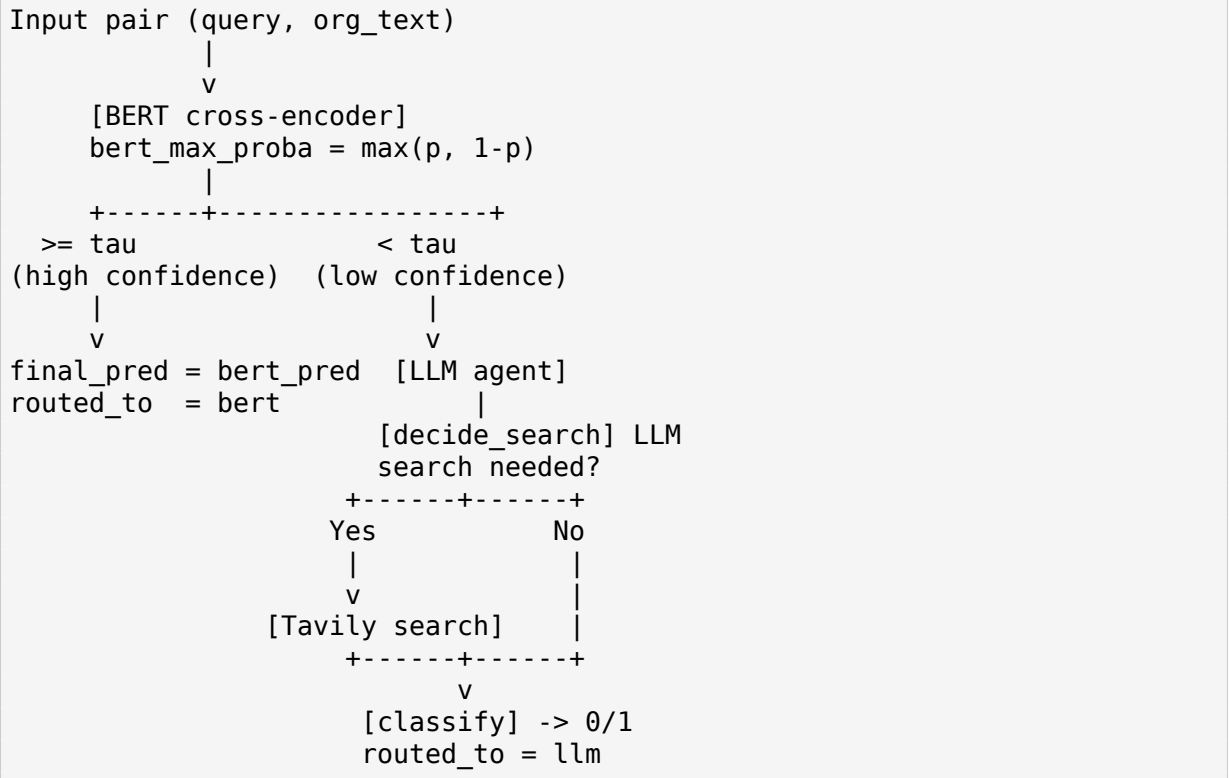


Figure 1: Confidence-gated hybrid pipeline.

3.2 Training Configuration

Table 3: Fine-tuning configuration (actual run on Google Colab, Tesla T4).

Parameter	Value
Base model	deepvk/RuModernBERT-base
Epochs requested / completed	5 / 2 (early stopping, patience = 1)
Train batch size	8
Eval batch size	16
Learning rate	2×10^{-5}
Weight decay	0.01
Warmup ratio	0.1
Mixed precision	fp16
Best model criterion	val accuracy
BERT_MAX_LENGTH	1024

Early stopping triggered after epoch 2 because val loss increased while accuracy did not improve.

Table 4: Training log per epoch.

Epoch	Train Loss	Val Loss	Accuracy	Macro-F1
1	0.5567	0.5050	0.7690	0.7683
2	0.4469	0.5488	0.7685	0.7681

3.3 Temperature Scaling

Calibration is performed via temperature scaling on val logits [3]. Temperature T is found by minimising NLL with LBFGS, then applied as $\hat{p} = \text{softmax}(\mathbf{z}/T)$. Since $\text{argmax}(\mathbf{z}/T) = \text{argmax}(\mathbf{z})$ for $T > 0$, the predicted class is unchanged; only the probabilities (and hence routing confidence) are affected.

Table 5: Temperature scaling results.

Metric	Before	After
Temperature T	1.000	1.1091
NLL	0.5050	0.4967
ECE	0.0564	0.0389

4 Pipeline Stages

Stages can be run through Python scripts (`scripts/run_stageN.py`) or directly in the corresponding Jupyter notebooks (`notebooks/stage*.ipynb`). Logic and artifacts are identical in both modes.

4.1 Stage 0 — EDA

Exploratory data analysis in `notebooks/eda.ipynb`: duplicate removal, justification of the binary formulation, split construction, and fixing of metrics and key project decisions. Key artifacts are stored in `data/processed/` and `reports/eda_reports/`.

4.2 Stage 1 — TF-IDF Baseline

TF-IDF (50k features, bigrams, sublinear TF) + Logistic Regression on `combined_text`. Serves as a lower bound; not part of the hybrid pipeline. Runtime: ≈ 20 min on CPU. Expected: val accuracy ≈ 0.63 – 0.65 , macro-F1 ≈ 0.62 – 0.64 .

4.3 Stage 2 — Fine-tune RuModernBERT + Temperature Scaling

Pipeline: load train/val/OOD $\rightarrow$ build `org_text` $\rightarrow$ validate labels and size $\rightarrow$ fine-tune $\rightarrow$ temperature scaling $\rightarrow$ save prediction parquets and `calibration.json`. GPU / Colab recommended.

4.4 Stage 3 — Error Analysis and Routing Threshold

This stage answers two questions: (1) at what `bert_max_proba` threshold to route an example to the LLM agent; (2) whether Tavily is needed or LLM alone is sufficient.

Threshold selection. The first threshold with coverage $\geq 70\%$ and accuracy on confident examples $\geq$ overall accuracy + 5 pp. If none is found, the default `CONFIDENCE_THRESHOLD = 0.75` is used.

Manual error taxonomy. About 96 rows are sampled from BERT errors on val: priority is given to high-confidence errors (`bert_max_proba` ≥ 0.75), stratified by TARGET class. Sample size follows Cochran’s formula [5]: $p \approx 0.30$, 95% confidence, ± 10 pp margin. Annotation categories: `requires_search`, `hard_semantic`, `fact_verification`, `label_noise`, `other`. If `searchable_share` $> 30\%$, then `agent_architecture` = `with_tavily`.

4.5 Stage 4 — Hybrid System on Val

Input: `val_merged_preds.parquet`. The LangGraph graph routes high-confidence examples to BERT and low-confidence ones through the LLM agent (optionally with Tavily). Outputs: `hybrid_val_preds.parquet` and `reports/stage4_agent/agent_metrics.json`.

4.6 Stage 5 — Final Evaluation on Eval (Locked)

Stage 5A runs BERT inference on `eval_baseline.parquet` (GPU/Colab) and saves `bert_eval_preds.parquet`. **Stage 5B** routes low-confidence eval examples through the LLM agent and assembles the final evaluation report. The threshold and temperature are not changed—they were fixed in Stages 2 and 3.

5 Metrics

We use **Accuracy** as the primary metric and **macro-F1** as the secondary metric. The choice of macro-F1 as the secondary metric is motivated by a slight class imbalance toward class 1 (52% in the balanced splits). Both metrics are fixed in Stage 0 and applied uniformly across all subsequent stages.

5.1 Accuracy

Accuracy measures the fraction of examples for which the predicted label matches the true label:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP are true positives, TN are true negatives, FP are false positives, and FN are false negatives.

5.2 Macro-F1

For each class $c \in \{0, 1\}$, precision P_c and recall R_c are defined as:

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c} \quad (2)$$

The per-class F1 score is the harmonic mean of precision and recall:

$$F1_c = \frac{2 P_c R_c}{P_c + R_c} \quad (3)$$

Macro-F1 is the unweighted average over all classes:

$$\text{macro-F1} = \frac{1}{|C|} \sum_{c \in C} F1_c = \frac{F1_0 + F1_1}{2} \quad (4)$$

Macro-averaging gives equal weight to both classes regardless of their support, which makes it sensitive to poor performance on the minority class.

6 Results

6.1 Overall Performance

Table 6: Summary results across splits and systems.

Split	System	Accuracy	Macro-F1	Note
Val ($n=4\,558$)	TF-IDF + LR	≈ 0.64	≈ 0.63	lower bound, CPU
Val ($n=4\,558$)	BERT-only	0.769	0.768	threshold tuned here
Val ($n=4\,558$)	Hybrid	0.774	0.773	
Eval ($n=4\,558$)	BERT-only	0.771	0.771	+0.2 pp vs. val
Eval ($n=4\,558$)	Hybrid	0.782	0.781	+0.8 pp vs. val

6.2 Routing and LLM Agent on Eval

Table 7: Routing statistics and LLM agent performance on the evaluation set.

Metric	Value
Confidence threshold τ	0.68
Low-confidence share	21.3% (971 / 4 558)
Agent accuracy on low-conf subset	0.618
BERT accuracy on same low-conf subset	0.573
Gain on low-confidence examples	+4.5 pp
Share of examples with Tavily search	13.5% of low-conf
LLM parse errors	0.7%
Total agent runtime	≈ 159 min

6.3 Error Matrix: Hybrid vs. BERT-only

The hybrid fixes 43 more examples than it breaks ($214 - 171 = 43$). The remaining potential lies in the “both wrong” category (18%), which requires improving the base model or expanding context.

Table 8: Error matrix comparing the hybrid system with BERT-only on eval ($n = 4527$).

Category	Examples	Share
Both correct	3 326	73.5%
Hybrid correct only	214	4.7%
BERT correct only	171	3.8%
Both wrong	816	18.0%

7 Known Limitations

LIMIT-01: Temperature scaling fitted on val. No separate calibration split: val is used for both model training and temperature selection. This introduces a small optimistic bias in T on eval.

LIMIT-02: Confidence threshold selected on val. No separate held-out set for threshold tuning. In practice, eval metrics matched or slightly exceeded val (hybrid accuracy 0.782 vs. 0.774), suggesting the threshold generalised well to unseen data. Stage 5 reveals the real picture.

LIMIT-03: Temperature scaling only. Platt scaling and isotonic regression were not used, as both risk overfitting on val (already used for model training). ECE after scaling is saved in `calibration.json`.

LIMIT-04: BERT_MAX_LENGTH = 1024. RuModernBERT supports up to 8 192 tokens. On OOM: reduce `batch_size` and/or add `gradient_accumulation_steps`.

LIMIT-05: LLM agent is non-deterministic. Results may vary slightly between runs. Setting `AGENT_USE_CACHE=true` and caching Tavily queries in `agent/search_cache/` reduces variability on repeated runs.

8 Reproducibility

- **BERT:** fix seed in `training_args` (Stage 2). Best checkpoint saved to `models/bert/best_c`
- **Calibration and threshold:** selected once on val, fixed in `calibration.json` and `comparison.json`. Not recomputed in Stages 4–5.
- **LLM agent:** enable `AGENT_USE_CACHE=true`—repeated Tavily and LLM requests return cached responses.
- **Splits:** created in `notebooks/eda.ipynb` and never modified afterwards.

9 Conclusion

We proposed a confidence-gated hybrid cascade for binary query–organisation relevance classification on Yandex Maps. RuModernBERT-base serves as the main classifier and handles the bulk of traffic; an LLM agent processes the low-confidence tail and, where the error taxonomy warrants it, calls a web-search tool.

On the locked evaluation set the hybrid achieves 78.2% accuracy and 0.781 macro-F1, which is +1.1 pp in accuracy over the BERT-only baseline and approximately +17 pp over TF-IDF + LR. The system fixes 43 more examples than it breaks relative to BERT-only. The remaining 18% “both wrong” category points to the clearest avenue for future improvement.

Next steps.

- Experiment with longer BERT_MAX_LENGTH (up to 8192) to capture full price-list and review text.
- Explore a dedicated calibration split to remove the optimistic bias from LIMIT-01 and LIMIT-02.
- Investigate query-level splitting (instead of row-level) to reduce data leakage between train and val.
- Fine-tune a larger RuModernBERT variant to reduce the “both wrong” residual.

References

- [1] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. Proceedings of NAACL-HLT 2019.
- [2] Nogueira, R., & Cho, K. (2019). *Passage Re-ranking with BERT*. arXiv:1901.04085.
- [3] Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). *On Calibration of Modern Neural Networks*. Proceedings of ICML 2017.
- [4] Ding, Y. et al. (2024). *Survey on Retrieval-Augmented Generation for Large Language Models*. arXiv:2404.14618.
- [5] Cochran, W. G. (1977). *Sampling Techniques* (3rd ed.). Wiley.